

Beyond Powerful Brains, VLAs, and WAMs

Rethinking the Cognitive and Sensorimotor
Foundations of Embodied Intelligence

Suhwan Choi

Outline

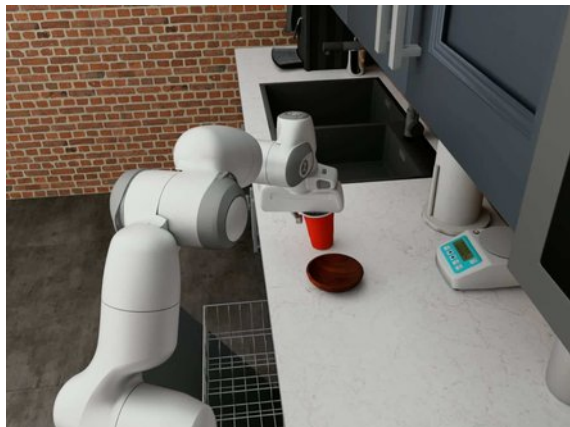
- 01 Cognitive and sensorimotor intelligence
- 02 Current approaches: capabilities and limits
- 03 Dual-clock systems: scaling both intelligences
- 04 Engineering multimodal scale

01

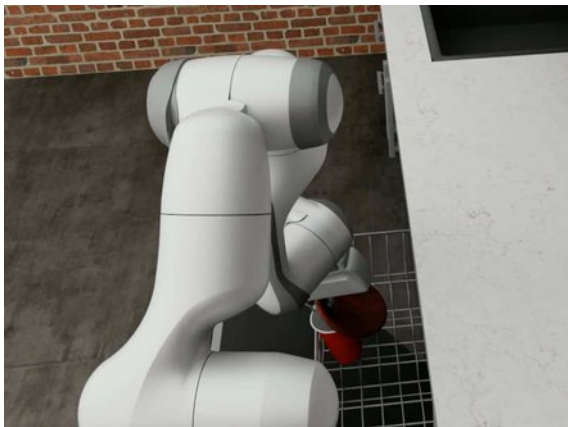
Cognitive and sensorimotor intelligence

What would it take to finish this task?

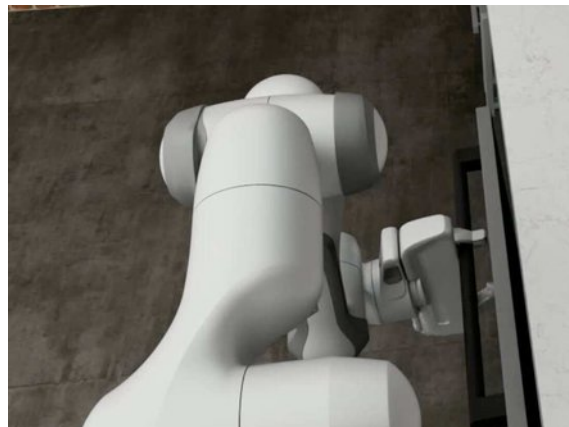
“Put the cup and bowl in the dishwasher.”



Pick up the objects



Move them into place



Close the door

Two kinds of intelligence, one task

Cognitive intelligence

Understand the situation.
Remember goals and events.
Reason, plan, and diagnose.

Sensorimotor intelligence

Track the body and objects.
Execute learned skills.
Adapt to physical feedback.

Strong reasoning. Skilled action. Both together.



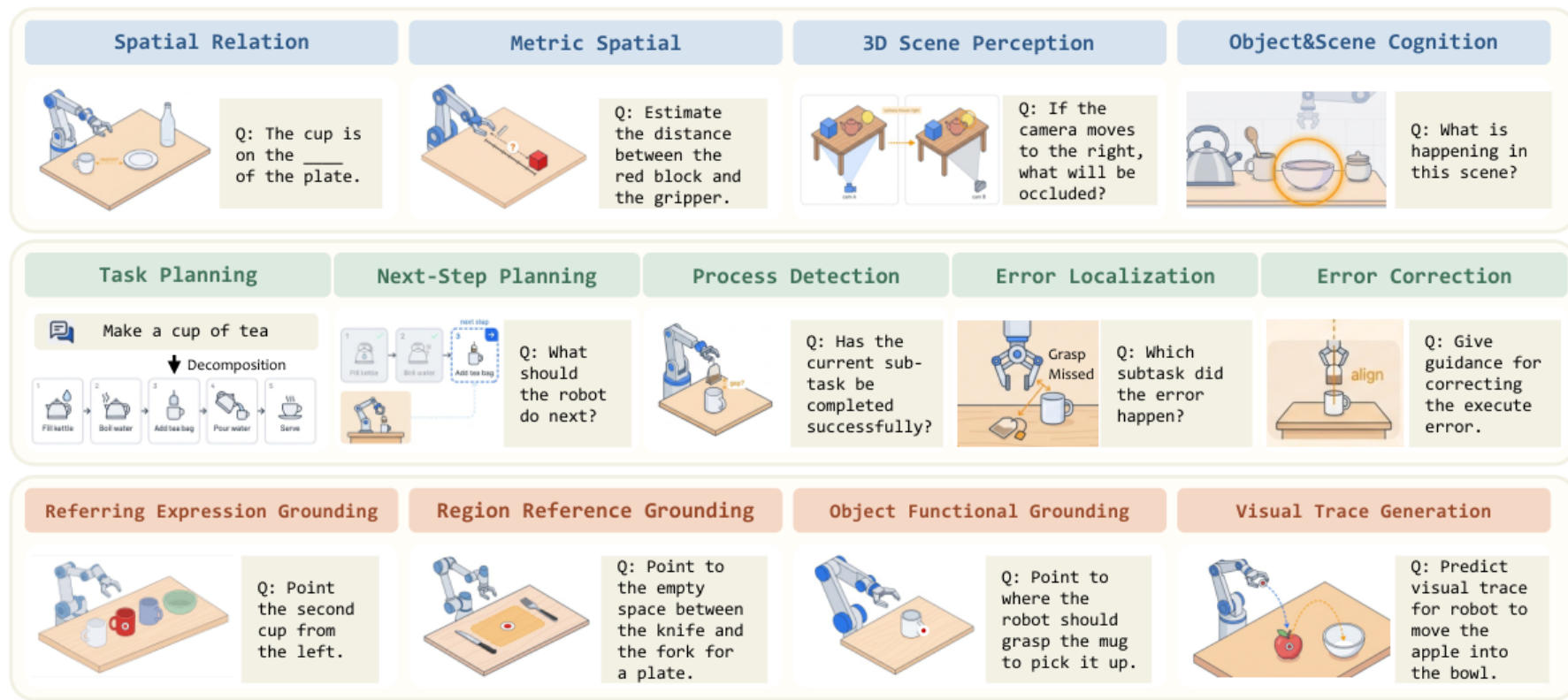
02

Current approaches: capabilities and limits

Different starting points, overlapping ambitions

Starting point	Learn to...	Typical output
Multimodal models	Interpret and reason	Answers, plans, grounding
VLA	Map observations to actions	Robot action sequences
WAM	Predict video and actions	Future frames + actions
Unified EFM	Cover reasoning and control	Multiple output types

What can a robot Brain contribute?



What can a powerful Brain add during execution?

TASK Place the mouse on the mouse pad, push the keyboard into the frame, put the figurine on the stand, place the alarm clock on the drawer, then open the drawer and put all remaining miscellaneous items inside.

HEAD VIEW **LEFT WRIST** **RIGHT WRIST**

GPT 6 Astra correction Decision 90 • Control 775 / 1000 • 31.00 s 1x simulation

CURRENT STATE • MODEL ASSESSMENT
Rotate the gripping direction across the stem.

DECISION RATIONALE
The hand rose and began rotating while the figurine stayed on the table. The return proposal leaves the fallen figurine unresolved. Continue the bounded rotation to change the gripping direction.

COMMAND DETAILS
Left ΔXYZ -0.0 +0.1 +0.0 cm
Right ΔXYZ +0.0 +0.0 +0.0 cm
Gripper opening: 0 closed • 1 open
Gripper opening L 1.00→1.00 R 1.00→1.00
Executed 5 / requested 5 steps
Targets are commands, not proof of arrival

Arrows: recorded EEF translation direction • L cyan / R amber • not proof of arrival GPT 6 Astra

[RoboDojo gallery](#)

Astra: task success, runtime, and cost

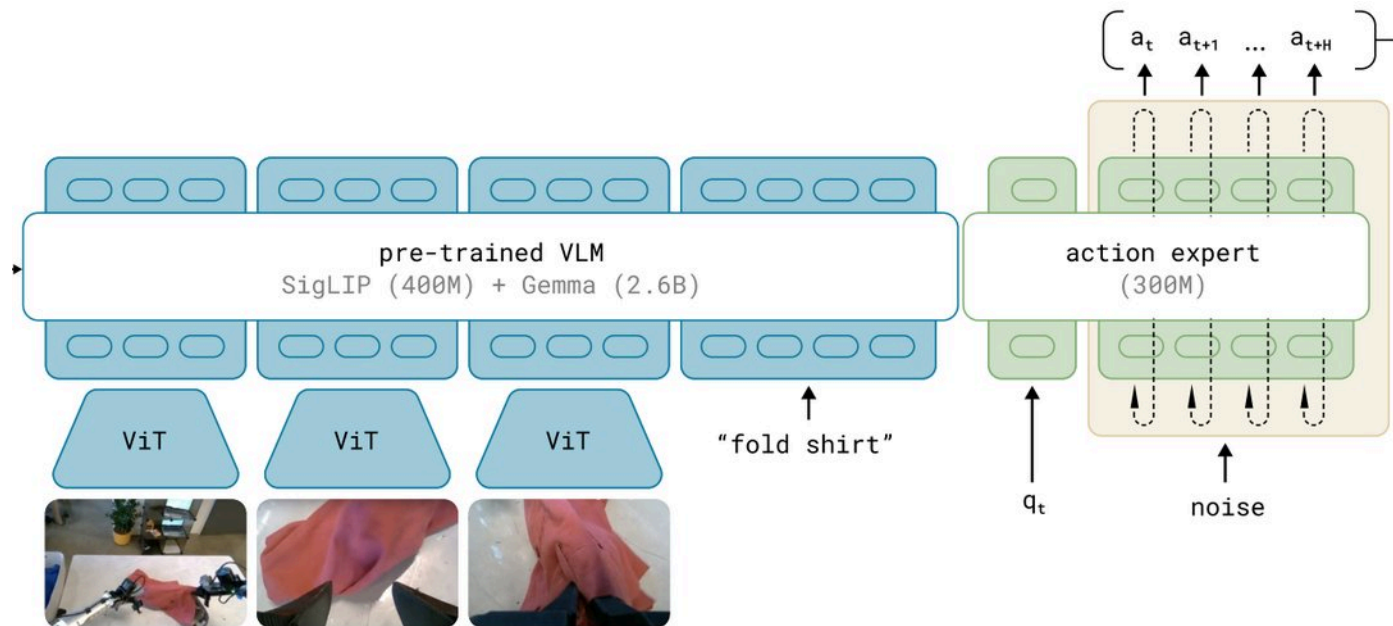
Robocurve (real robot)

Task	Success	Mean time	Est. cost / trial
Block into bowl	19/20	2.5 min	\$0.94
Puzzle insertion	2/20	3.4 min	\$1.36

RoboDojo (simulation)

Policy	Success	Est. cost / episode
Astra direct	13/50	~\$29
π0.5 + Astra	24/50	~\$17

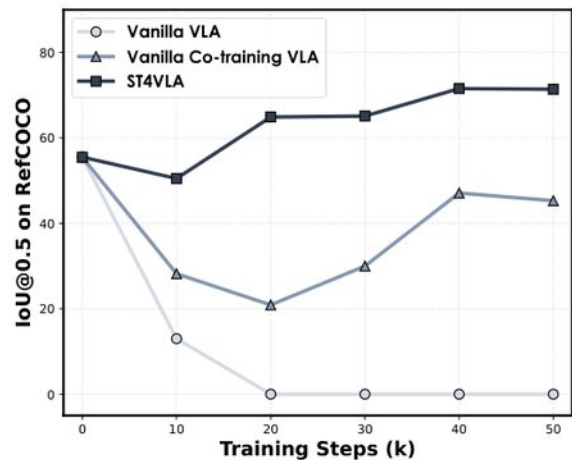
VLA: from observations to robot actions



VLM $\rightarrow$ action expert $\rightarrow$ action chunk

Action training can erode pretrained VLM capabilities

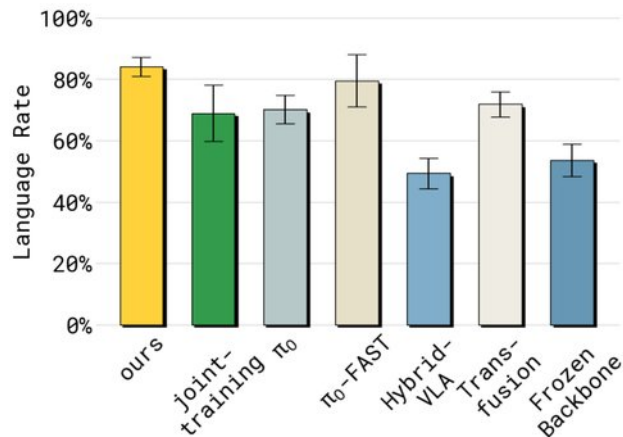
Spatial grounding



(a) Perception Performance

Vanilla VLA: $\approx 55\% \rightarrow \approx 0\%$

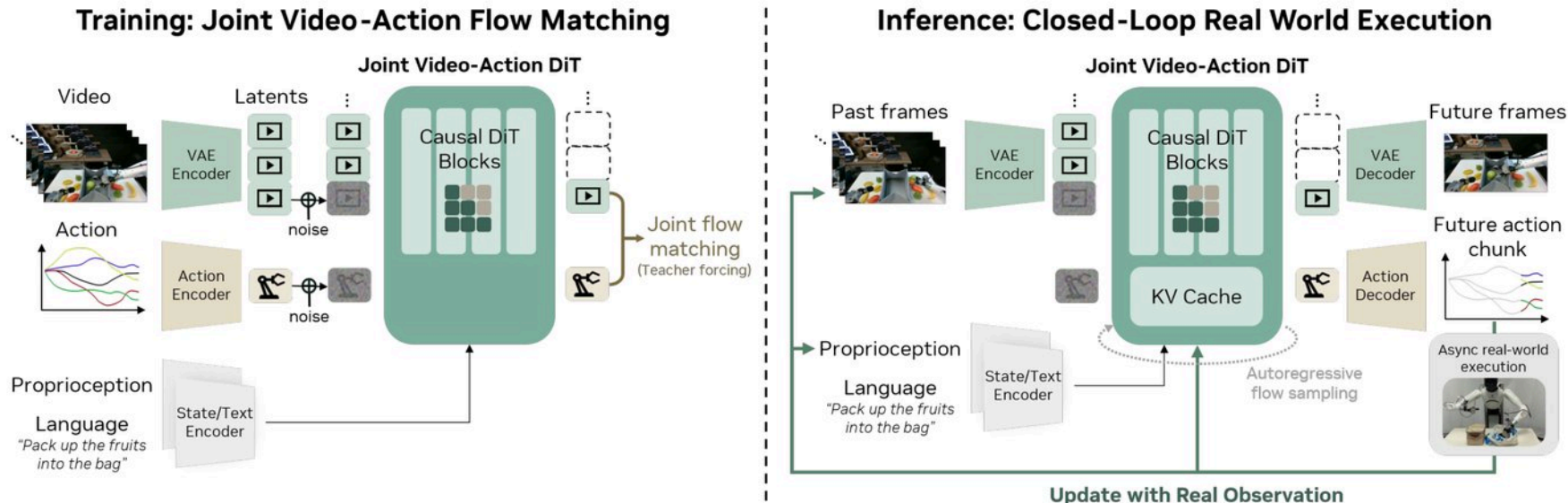
Language following



(b) Language following

Joint $\approx 70\%$ / Insulated $\approx 85\%$

WAM: learning actions with future prediction



Why make a video generator the foundation?

Cognitive intelligence

Why start here rather than an MLLM?

Sensorimotor intelligence

What justifies the additional inference cost?

EFM: Embodied Foundation Models



[Embodied-R1.5](#)

Embodied reasoning
and action adaptation

[Isaac 0.5](#)

Video understanding,
reasoning, and control

[Cosmos 3](#)

Understanding, generation,
and action

Can one model retain strong reasoning and strong control?

Building foundations directly from physical data

System	Foundation training	Reported outcome
Generalist GEN-1.5	Ground-up pretraining on physical experience	One-shot task learning
Sunday ACT-1	Human glove data with embodiment alignment	Household manipulation
Reward AI OM-1	Single-stage learning from human interaction	Cross-embodiment control

03

Dual-clock systems: scaling both intelligences

Smaller policies. Larger foundations again?

[RT-2](#)

2023

55B

Total parameters

[OpenVLA](#)

2024

→ **7B**

[X-VLA](#)

2025

→ **0.9B**

[Isaac 0.5](#)

2026

→ **36B**

2.5B active per token

How do we scale intelligence without slowing action?

Scale the Brain. Keep sensorimotor inference fast.

Cognitive intelligence

- More capacity
- Longer context
- More reasoning compute

Sensorimotor intelligence

- Fresh observations
- Low latency
- Predictable compute

Scale cognition without making every action wait for it.

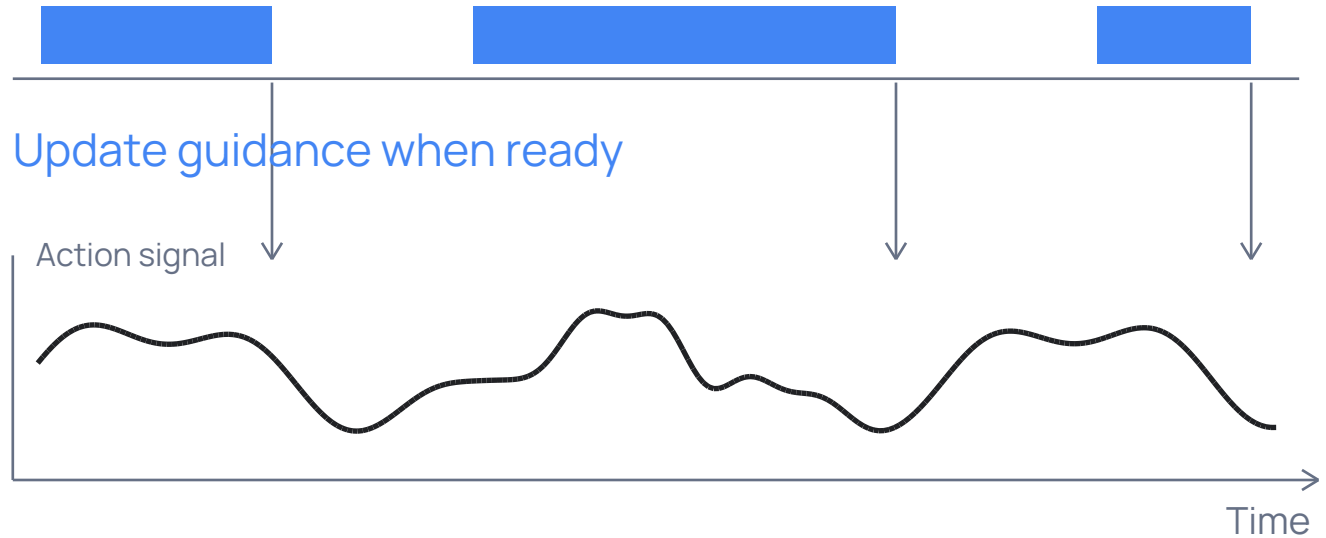
Think slow, act fast: a dual-clock system

Cognition

System 2

Sensorimotor

System 1



System 1 predicts actions; System 0 executes them.

Separate components, clocks, and reaction time.

Two components $\neq$ two independent clocks

[GR00T N1.7](#)

1 policy call = 1 backbone call + 1 action-head call (synchronized)

Inference interval $\neq$ reaction time

Inference interval

One model invocation $\rightarrow$ the next

Reaction time

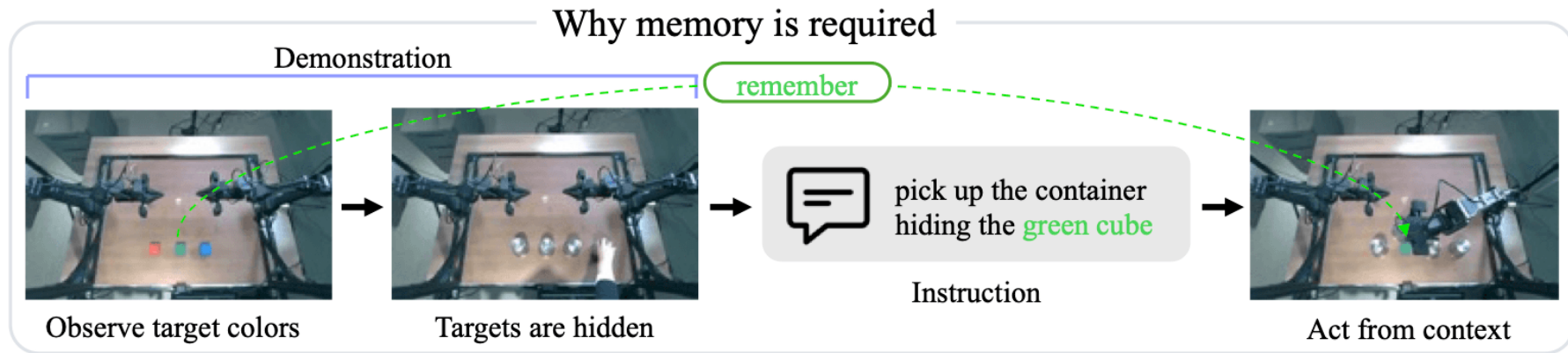
A change in the world $\rightarrow$ a responsive action

[1,000 Hz command playback can still take \$\geq 0.5\$ s to react.](#)

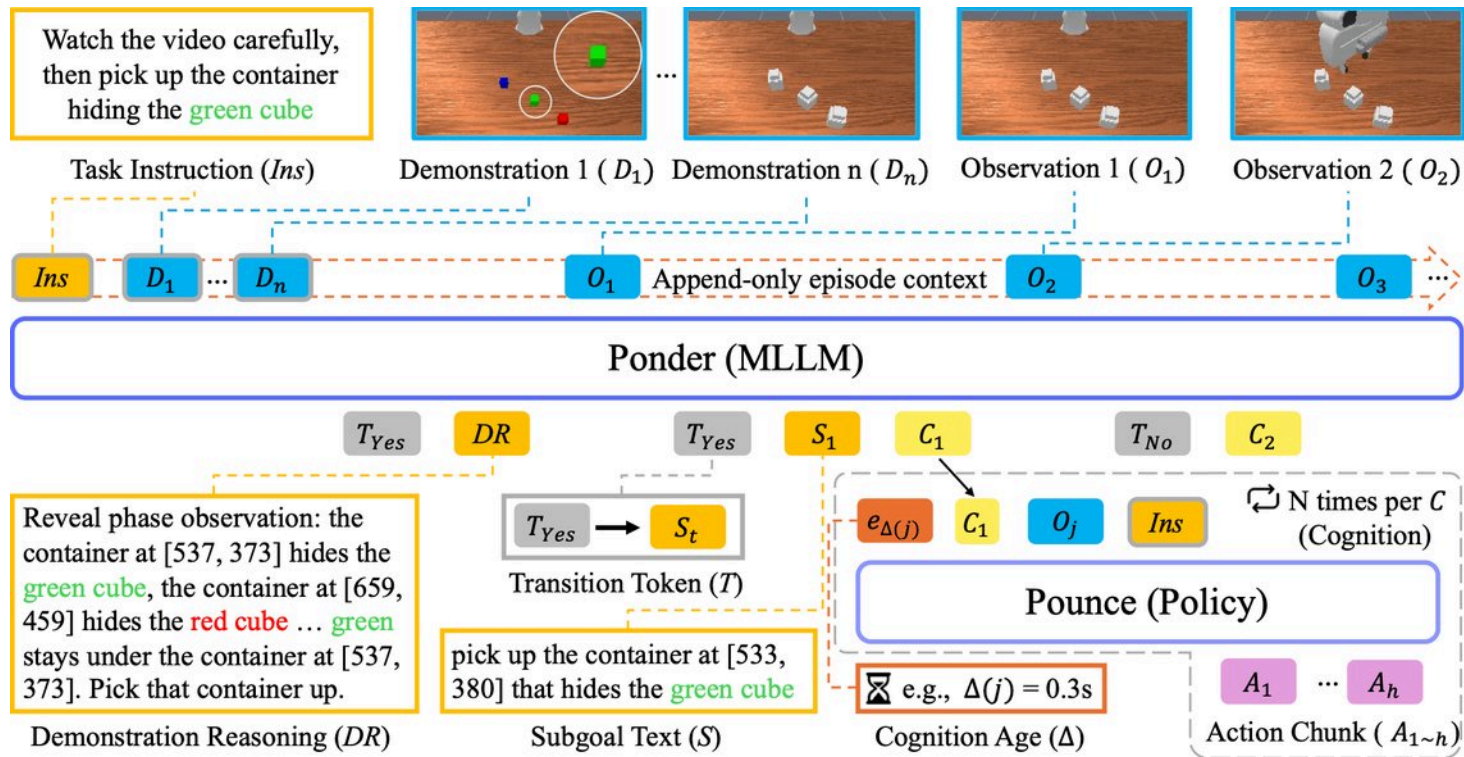
PonderPounce: learning to act from memory

Suhwan Choi et al.

Can remembered context guide present action?



PonderPounce: connecting context to action

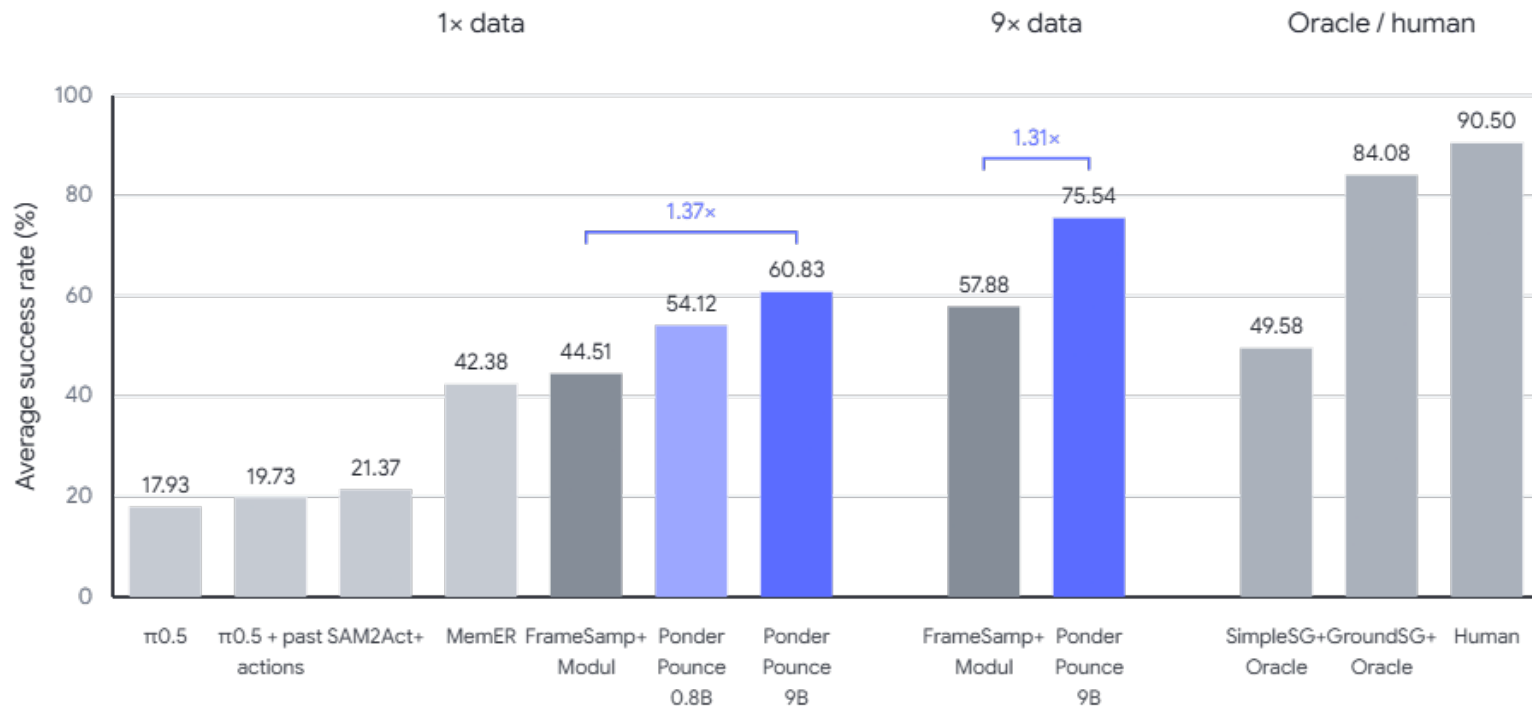


PonderPounce: separate the inference paths

Inference path	Latency (H100, p50)
Pounce: action prediction	25 ms
Ponder: cognition refresh	78 ms
Ponder: cognition + 45-token subgoal	820 ms

Pounce acts while Ponder thinks.

PonderPounce: performance and scaling



PonderPounce is a step, not the finished system.

Strong gains, but below the level we expected.

Memoryless π0.5 1× data	FrameSamp+ Modul (9× data)	PonderPounce 9× data	GroundSG+Oracle Reference	Human Reference
17.93%	57.88%	75.54%	84.08%	90.50%

We expected performance near the 84.08% oracle reference.

Suspected bottlenecks

- The Ponder–Pounce interface
- Pounce itself

04

Engineering multimodal scale

Data engineering has four distinct problems

Recording

Observations and actions arrive at different rates.

Timestamping and stream synchronization

Storing

Large media collections must stay accessible.

Compression, sharding, and indexing

Loading

Training batches must keep up with the GPU.

Sampling, decoding, and sequence packing

Infra

Many training nodes compete for shared data.

Storage capacity, network bandwidth, and fault tolerance

Multimodal foundations need a data system.

Team	Stack	Purpose
Cosmos 3	SILA , Lance , Ray	Curate and query at corpus scale
Isaac 0.5	mHarmony	Unify samples Keep GPUs supplied
WoRV	blobpack MediaRef CephFS	Package media into shards Reference images and video frames Serve shared files across nodes

Open-source tools for multimodal training

Suhwan Choi

[blobpack](#)

Portable ZIP shards for media.
Fast random and sequential reads.

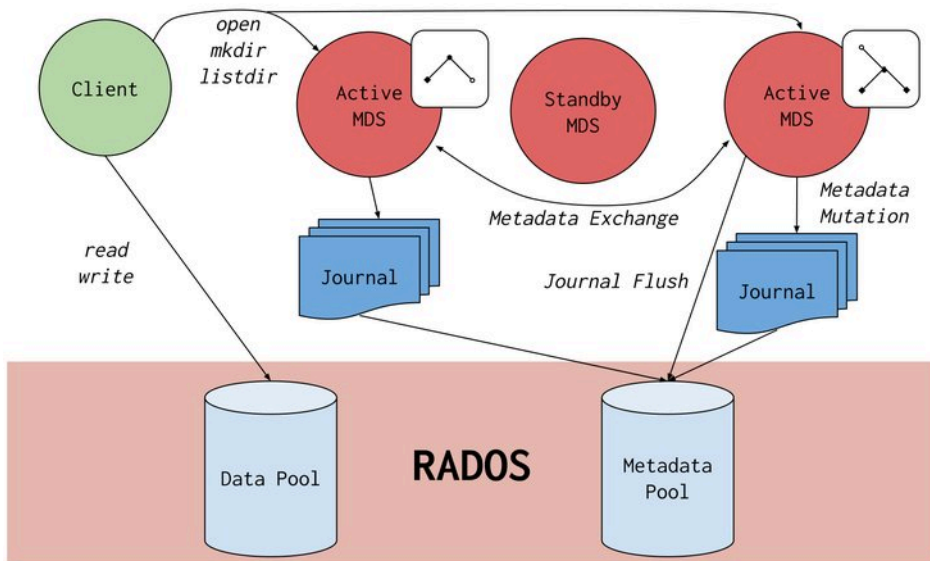
[MediaRef](#)

Portable image and video-frame references.
One schema with batch decoding.

Shared files on distributed storage

[CephFS](#): shared files across training nodes.

Native Linux kernel client. Supported in Red Hat Ceph Storage.



Metadata operations

Via MDS

File reads / writes

Direct to RADOS

Data and metadata
persist in RADOS.

Cepheid: built and tuned for training

Suhwan Choi & Jiyong Yoon

8 storage nodes / 64 NVMe / 492 TB raw / 200 GbE

EC 6+2 bulk pool / 3-replica hot pool

Aggregate across 12 training clients

162 GB/s

Sequential read

2.23M IOPS

4K random read

227 TB

Currently stored

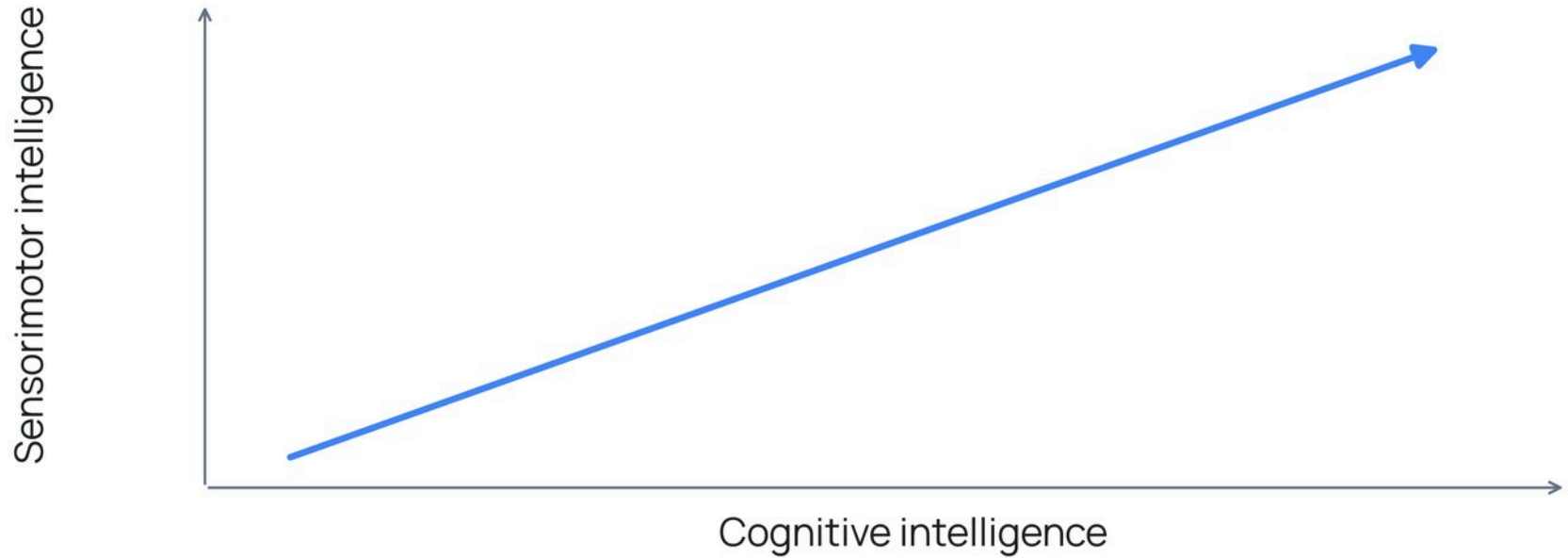
380 TB

Cumulative reads

377 TB

Cumulative writes

Let's scale both.



Questions

References: models and systems

[\$\pi\$ 0 – Vision-Language-Action Flow Model](#)

[DreamZero – World Action Models](#)

[Embodied-R1.5](#)

[Isaac 0.5 – Perceptron](#)

[GPT-as-Policy – independent Astra evaluation](#)

[PonderPounce](#)

[Cosmos 3](#)

References: evaluation and control

[MMMU](#)

[BabyVision](#)

[ERQA](#)

[RoboCasa365](#)

[RoboDojo](#)

[Revisiting Open-Loop Execution](#)

[Knowledge Insulation](#)

[VLASER](#)

References: physical-data-first models

[Sunday ACT-1](#)

[Generalist GEN-1.5](#)

[Reward AI OM-1](#)

References: systems and engineering

[RT-2](#)

[OpenVLA](#)

[X-VLA](#)

[GR00T N1.7 implementation](#)

[Cosmos 3 – infrastructure](#)

[blobpack](#)

[MediaRef](#)

Appendix

Towards reactive control



Dispense exactly one pill.

When a pill drops...

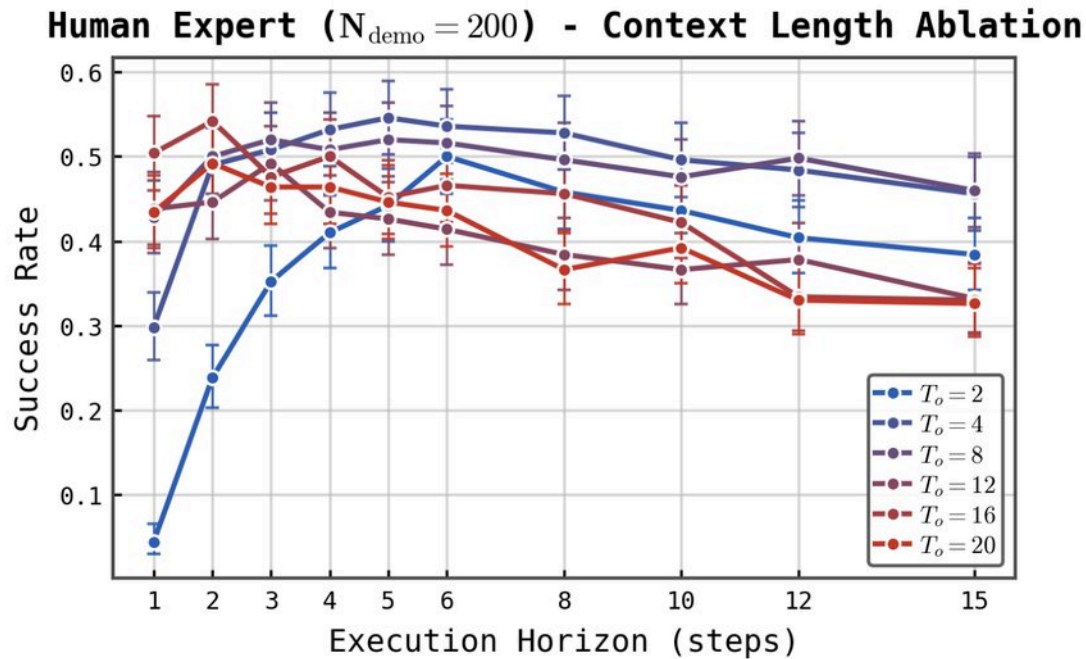
Open-loop execution

The queued actions continue.

Reactive execution

Fresh feedback changes the next action.

Reactive control needs memory, too



Recent history →
state estimation.

Longer history:
best success near
1-2 execution steps.